

IPL DATA ANALYTICS PROJECT REPORT

Exploratory Analysis of Indian Premier League Cricket
Using PostgreSQL and Power BI

PROJECT TYPE	Data Analytics and Business Intelligence
TECHNOLOGIES	Python, Pandas, PostgreSQL, SQL, Microsoft Power BI
DATASET	IPL Ball-by-Ball Match Data (1,243 matches)
SEASONS COVERED	2007/08 to 2026 (as labelled in the dataset)
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Table of Contents

Table of Contents	2
Abstract	4
1. Introduction	5
2. Problem Statement	5
3. Project Objectives	5
4. Dataset Description	6
5. Tools and Technologies	7
5.1 Python and Pandas	7
5.2 PostgreSQL and SQL	7
5.3 Microsoft Power BI	7
6. Data Preparation and Methodology	7
6.1 Data loading and inspection	7
6.2 Data cleaning	7
6.3 Database loading and SQL analysis	8
6.4 Power BI visualisation	8
7. SQL Analytics	8
7.1 Q1 - Scoring across seasons	9
7.2 Q2 - Team scoring consistency	9
7.3 Q3 - Chasing success	10
7.4 Q4 - Does the toss matter?	10
7.5 Q5 - Powerplay aggressiveness	11
7.6 Q6 - Powerplay to death-over acceleration	11
7.7 Q7 - Batting strike rates (minimum 500 balls)	11
7.8 Q8 - Bowlers who create dot-ball pressure	12
7.9 Q9 - Batting-friendly venues	12
7.10 Q10 - Overall team batting profile	13
8. Power BI Dashboard	14
8.1 Page 1 - IPL Overview	14
8.2 Page 2 - Season & Scoring Trends	14
8.3 Page 3 - Team Performance	15
8.4 Page 4 - Player Performance	16
8.5 Page 5 - Venue & Match Insights	17

9. Key Observations	18
10. Limitations	18
11. Conclusion	19
12. Future Scope	19
References	19
Appendix A - SQL Queries	20

Abstract

The Indian Premier League (IPL) generates a large amount of cricket data covering individual deliveries, player performances, team results, match conditions, venues and scoring patterns. Analysing this data helps identify historical trends, evaluate player and team performance, and understand factors associated with match outcomes.

This project presents an IPL analytics solution using PostgreSQL for structured analysis and Microsoft Power BI for interactive visualisation. Ball-by-ball data covering 1,243 matches is used to examine seasonal scoring trends, team performance, batting and bowling statistics, venue characteristics and toss-related patterns.

Ten SQL queries perform the aggregations, comparisons and performance calculations. The results are presented through a five-page Power BI dashboard: IPL Overview, Season & Scoring Trends, Team Performance, Player Performance, and Venue & Match Insights.

Key results include a steady rise in scoring (average innings score of about 141 in 2009 to about 185 in 2026), a toss-winner win rate of only 50.52%, and large differences in batting profile between teams and venues. The project shows how database querying and business-intelligence tools turn raw cricket data into an organised, explorable analytical dashboard.

1. Introduction

Cricket analytics has become an important part of understanding player performance, team strategies and match conditions. The Indian Premier League is particularly suitable for analysis because it contains numerous matches, multiple teams, diverse venues and detailed ball-by-ball records.

Traditional scorecards provide individual match statistics, but analysing many seasons manually makes long-term patterns hard to see. Data analytics tools process these records efficiently and present the results visually.

This project analyses IPL data using PostgreSQL and Power BI. PostgreSQL is used to query and aggregate the data, while Power BI provides interactive charts, KPI cards and filters. The dashboard lets users examine the IPL at several levels: overall tournament statistics, season-wise trends, team comparisons, player rankings and venue-level analysis.

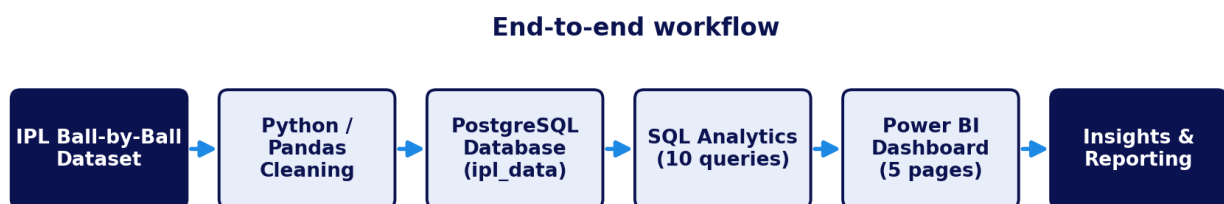


Figure 1. Overall workflow of the IPL data analytics project: dataset, Python cleaning, PostgreSQL storage, SQL analytics, Power BI dashboard and reporting.

2. Problem Statement

IPL ball-by-ball data contains detailed information about deliveries, runs, wickets, players, teams, seasons, venues and match outcomes. Raw records alone, however, do not give an easily understandable overview of tournament performance. The problem addressed here is the need to organise, analyse and visualise this data to identify useful statistical patterns.

The project aims to answer the following questions:

- How have scoring patterns changed across IPL seasons?
- Which teams have recorded stronger batting performances?
- Which players have accumulated the most runs and wickets?
- How do strike rates and bowling economy rates compare?
- Which venues have hosted the most matches or recorded higher average match totals?
- What patterns can be observed in toss decisions and match outcomes?

3. Project Objectives

1. Clean and prepare IPL ball-by-ball data for analysis.
2. Store and query structured cricket data using PostgreSQL.
3. Perform analytical SQL queries using aggregations, conditional calculations and common table expressions.

4. Analyse seasonal scoring patterns and run rates.
5. Compare team and player performance using relevant statistical measures.
6. Study venue characteristics and toss-related patterns.
7. Develop an interactive five-page Power BI dashboard.
8. Present the analysis through clear visualisations and a consolidated report.

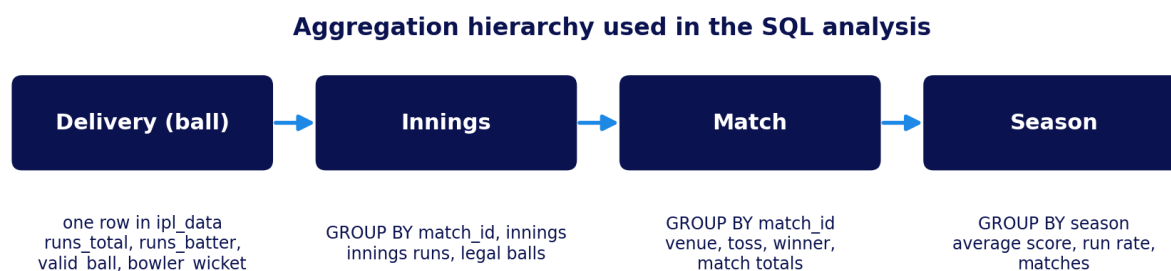
4. Dataset Description

The project uses an IPL ball-by-ball dataset with delivery-level records and match-related information: match identifiers, innings, overs, batters, bowlers, batting and bowling teams, runs, extras, wickets, seasons, venues, toss decisions and match winners. The dataset covers 1,243 distinct matches. Key fields are described in Table 1.

Table 1. Key fields in the ipl_data table

Field	Description
match_id	Identifier used to group deliveries belonging to a match
season	IPL season associated with a record
innings	Innings number
over	Over number (0-19)
batter / bowler	Batter and bowler associated with a delivery
batting_team / bowling_team	Team batting / bowling during the delivery
runs_batter	Runs credited to the batter
runs_total	Total runs recorded for the delivery
valid_ball	Indicator for legal deliveries
bowler_wicket	Indicator used in the bowling-wicket calculation
venue	Match venue
toss_winner / toss_decision	Team that won the toss and its decision (bat or field)
match_won_by	Team recorded as the match winner
win_outcome	Winning margin or outcome description

The fields are used to group deliveries into innings and matches before the required statistics are calculated (Figure 2). The season labels used by the dataset are 2007/08, 2009, 2009/10, 2011 to 2019, 2020/21 and 2021 to 2026, and they are shown exactly as stored.



Statistics are computed at the correct level (e.g., innings totals first, then averaged across innings).

Figure 2. Aggregation hierarchy: delivery-level records are aggregated to innings, match and season level.

5. Tools and Technologies

5.1 Python and Pandas

Python and Pandas were used to load the dataset, inspect its structure, identify duplicate records, handle data types, clean column names, remove stray whitespace from text values and export the cleaned dataset.

5.2 PostgreSQL and SQL

PostgreSQL stored the prepared data in a single analysis table named `ipl_data`. SQL aggregated ball-by-ball records into match-, innings-, team-, player- and venue-level statistics using GROUP BY, COUNT, SUM, AVG, STDDEV, CASE, FILTER, HAVING and common table expressions (CTEs).

5.3 Microsoft Power BI

Power BI was used to build the interactive dashboard, which contains KPI cards, line, bar and area charts, donut charts and slicers for season, venue, team, batter and bowler.

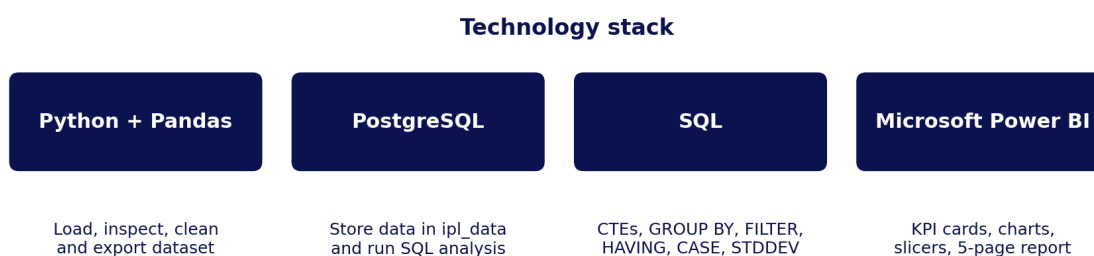


Figure 3. Technologies used in the IPL analytics workflow.

6. Data Preparation and Methodology

6.1 Data loading and inspection

The dataset was loaded into Python with Pandas. Initial inspection covered the number of records, columns, data types and the presence of missing or duplicate values.

6.2 Data cleaning

- Checking for and removing duplicate rows.

- Standardising column names to lowercase with underscores.
- Removing leading and trailing spaces from text values.
- Converting the date field to a date type and selected fields to numeric types.
- Re-checking missing values and duplicates after cleaning.

6.3 Database loading and SQL analysis

The cleaned file was loaded into PostgreSQL, where `ipl_data` served as the source for ten analytical queries covering seasonal scoring, team consistency, chasing, toss outcomes, Powerplay scoring, death-over acceleration, batting strike rates, bowling dot-ball pressure, venue scoring and overall team batting profile.

6.4 Power BI visualisation

The analytical data was used to create a five-page Power BI report. Measures were created for KPIs and comparisons, and slicers were added for interactive filtering.

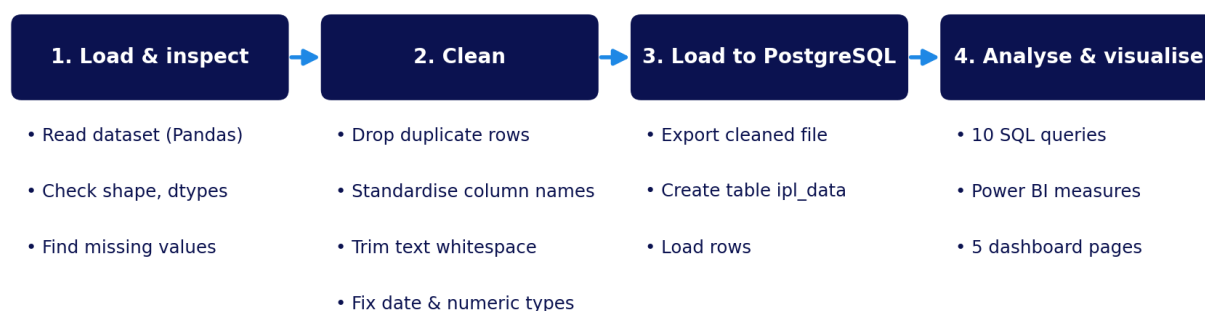


Figure 4. Data preparation, database loading, analysis and visualisation process.

7. SQL Analytics

The PostgreSQL analysis addresses ten questions (Table 2). Statistics are calculated at the correct level: for example, seasonal scoring groups deliveries into innings first, and venue scoring groups runs by match before averaging. Minimum-sample conditions (for example at least 20 innings, 10 matches or 500 legal balls) prevent rankings from being driven by very small samples. The complete SQL is in Appendix A.

Table 2. The ten analytical questions

No.	Analytical question	Main metrics
1	How has IPL scoring changed across seasons?	Average innings score, run rate, matches
2	Which teams are most consistent at scoring runs?	Average score, score variation, min and max scores
3	Which teams are most successful when chasing?	Chases, successful chases, success rate
4	Does winning the toss increase the chance of winning?	Toss-winner wins, toss-win percentage
5	Which teams score most aggressively in the Powerplay?	Average Powerplay runs, Powerplay run rate
6	Which teams accelerate most from Powerplay to death overs?	Powerplay run rate, death-over run rate, acceleration
7	Which batters have the best strike rates (min. 500 balls)?	Runs, balls faced, strike rate
8	Which bowlers create the most dot-ball pressure?	Legal balls, dot balls, dot-ball %
9	Which venues are most batting-friendly?	Average match runs, score variation
10	Which teams have the strongest overall batting profile?	Runs per match, fours, sixes, run rate, dot-ball %

7.1 Q1 - Scoring across seasons

Average innings score moved from 154.63 in 2007/08 to a low of 140.97 in 2009, then recovered, reaching 165.84 in 2018. Run rate shows the same pattern (8.31, 7.49 and 8.65 respectively). Later seasons are shown on the dashboard in Section 8.2.

	season text	matches bigint	average_innings_score numeric	run_rate numeric
1	2007/08	58	154.63	8.31
2	2009	57	140.97	7.49
3	2009/10	60	154.78	8.13
4	2011	73	145.89	7.73
5	2012	74	151.71	7.83
6	2013	76	144.88	7.69
7	2014	60	155.17	8.20
8	2015	59	154.23	8.37
9	2016	60	157.18	8.31
10	2017	59	156.55	8.41
11	2018	60	165.84	8.65

Figure 5. Query 1 result (first 11 seasons shown): matches, average innings score and run rate per season.

7.2 Q2 - Team scoring consistency

Royal Challengers Bengaluru have the highest average score (187.83 over 46 innings), followed by Gujarat Titans (178.48) and Punjab Kings (178.26). Among the rows shown, Gujarat Titans (32.25) and Chennai Super Kings (32.40) have the lowest score variation, meaning the most consistent totals. Delhi

Capitals (42.75) and Sunrisers Hyderabad (42.86) are the most variable; Sunrisers also hold the highest score in the table (287).

	batting_team text	innings_played bigint	average_score numeric	score_variation numeric	lowest_score numeric	highest_score numeric
1	Royal Challengers Bengal...	46	187.83	38.14	77	262
2	Gujarat Titans	77	178.48	32.25	89	233
3	Punjab Kings	87	178.26	37.21	98	265
4	Lucknow Super Giants	73	174.71	38.67	1	257
5	Chennai Super Kings	266	163.91	32.40	9	246
6	Delhi Capitals	123	162.27	42.75	3	264
7	Sunrisers Hyderabad	214	161.87	42.86	2	287
8	Mumbai Indians	296	160.95	37.34	5	247
9	Rajasthan Royals	254	159.20	39.30	6	243
10	Gujarat Lions	31	156.84	37.17	6	208
11	Royal Challengers Bangal...	243	155.11	38.74	2	263

Figure 6. Query 2 result: average score, score variation, lowest and highest innings totals for teams with at least 20 innings.

7.3 Q3 - Chasing success

Gujarat Lions (75.00% from 16 chases) and Royal Challengers Bengaluru (72.73% from 22) lead, but both are small samples. Among teams with larger samples, Chennai Super Kings succeed in 59.50% of 121 chases, Mumbai Indians 55.32% of 141 and Kolkata Knight Riders 55.17% of 145.

	chasing_team text	chases bigint	successful_chases bigint	chase_success_rate numeric
1	Gujarat Lions	16	12	75.00
2	Royal Challengers Bengal...	22	16	72.73
3	Gujarat Titans	43	27	62.79
4	Punjab Kings	40	24	60.00
5	Chennai Super Kings	121	72	59.50
6	Mumbai Indians	141	78	55.32
7	Kolkata Knight Riders	145	80	55.17
8	Rajasthan Royals	133	73	54.89
9	Delhi Capitals	67	35	52.24
10	Royal Challengers Bangal...	119	60	50.42
11	Sunrisers Hyderabad	98	48	48.98

Figure 7. Query 3 result: chases, successful chases and chase success rate (teams with at least 10 chases).

7.4 Q4 - Does the toss matter?

Across 1,243 matches the toss winner won 628 (50.52%). The edge over a coin-flip split is only about half a percentage point, so the toss on its own gives no meaningful historical advantage.

	total_matches bigint	toss_winner_wins bigint	toss_win_percentage numeric
1	1243	628	50.52

Figure 8. Query 4 result: matches won by the toss winner.

7.5 Q5 - Powerplay aggressiveness

Royal Challengers Bengaluru are the most aggressive Powerplay side (59.96 average runs, 9.99 per over), followed by Punjab Kings (54.83, 9.14). Chennai Super Kings (7.90) and Mumbai Indians (7.97) have the lowest Powerplay run rates among the rows shown.

	batting_team text	innings bigint	average_powerplay_runs numeric	powerplay_run_rate numeric
1	Royal Challengers Bengal...	46	59.96	9.99
2	Punjab Kings	87	54.83	9.14
3	Gujarat Titans	77	51.92	8.65
4	Delhi Capitals	119	51.98	8.63
5	Lucknow Super Giants	72	51.79	8.62
6	Gujarat Lions	30	51.97	8.61
7	Sunrisers Hyderabad	210	50.93	8.48
8	Rajasthan Royals	250	49.08	8.18
9	Kolkata Knight Riders	278	48.16	8.05
10	Mumbai Indians	291	47.90	7.97
11	Chennai Super Kings	265	47.40	7.90

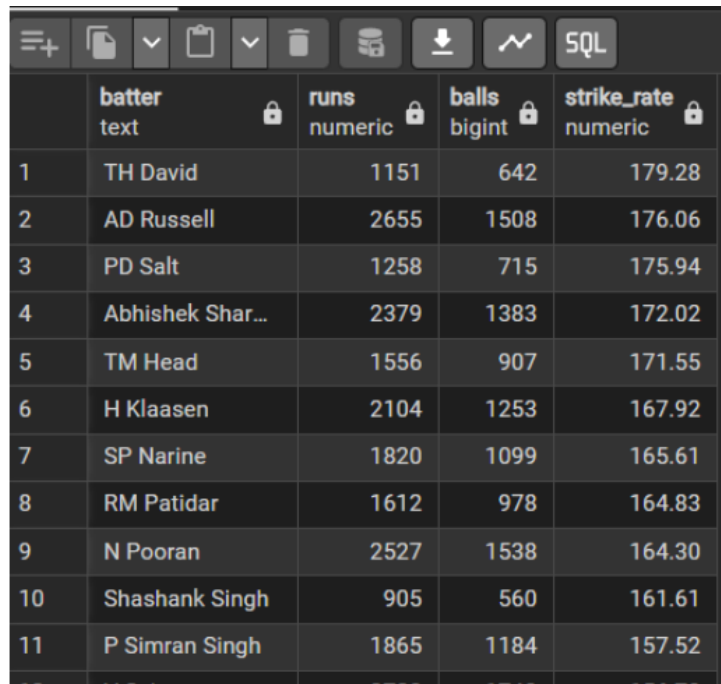
Figure 9. Query 5 result: average Powerplay runs and Powerplay run rate per team.

7.6 Q6 - Powerplay to death-over acceleration

Query 6 compares each team's run rate in overs 0-5 with overs 16-19 and reports the difference as acceleration. The query is given in Appendix A.

7.7 Q7 - Batting strike rates (minimum 500 balls)

TH David (179.28), AD Russell (176.06) and PD Salt (175.94) top the list, followed by Abhishek Sharma (172.02) and TM Head (171.55). The 500-ball filter removes players whose strike rates are inflated by a handful of deliveries.

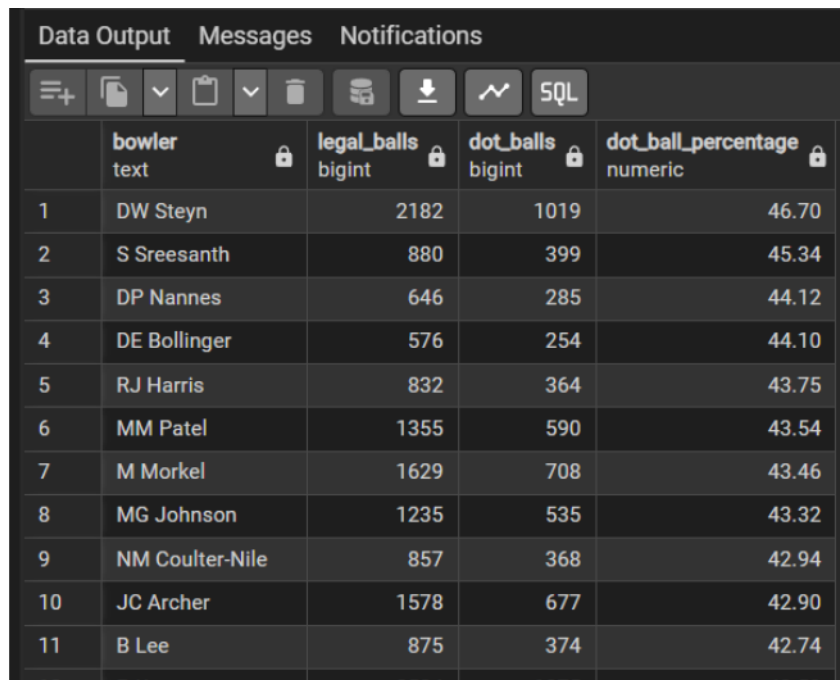


	batter text	runs numeric	balls bigint	strike_rate numeric
1	TH David	1151	642	179.28
2	AD Russell	2655	1508	176.06
3	PD Salt	1258	715	175.94
4	Abhishek Shar...	2379	1383	172.02
5	TM Head	1556	907	171.55
6	H Klaasen	2104	1253	167.92
7	SP Narine	1820	1099	165.61
8	RM Patidar	1612	978	164.83
9	N Pooran	2527	1538	164.30
10	Shashank Singh	905	560	161.61
11	P Simran Singh	1865	1184	157.52

Figure 10. Query 7 result: top batters by strike rate among those who faced at least 500 legal balls.

7.8 Q8 - Bowlers who create dot-ball pressure

DW Steyn leads with 46.70% dot balls over 2,182 legal balls, followed by S Sreesanth (45.34%), DP Nannes (44.12%) and DE Bollinger (44.10%). Steyn's figure is notable because it comes from the largest sample in the top ten.



	bowler text	legal_balls bigint	dot_balls bigint	dot_ball_percentage numeric
1	DW Steyn	2182	1019	46.70
2	S Sreesanth	880	399	45.34
3	DP Nannes	646	285	44.12
4	DE Bollinger	576	254	44.10
5	RJ Harris	832	364	43.75
6	MM Patel	1355	590	43.54
7	M Morkel	1629	708	43.46
8	MG Johnson	1235	535	43.32
9	NM Coulter-Nile	857	368	42.94
10	JC Archer	1578	677	42.90
11	B Lee	875	374	42.74

Figure 11. Query 8 result: dot-ball percentage for bowlers with at least 500 legal balls.

7.9 Q9 - Batting-friendly venues

Arun Jaitley Stadium, Delhi has the highest average match total (375.13 runs over 30 matches), followed by HPCA Stadium, Dharamsala (374.00, 10 matches) and Sawai Mansingh Stadium, Jaipur (372.81). Wankhede Stadium, with 59 matches, averages 353.44. Score variation at Eden Gardens

(90.52) and Dharamsala (96.13) is high, so totals there are less predictable.

	venue text	matches bigint	average_match_runs numeric	score_variation numeric
1	Arun Jaitley Stadium, Delhi	30	375.13	74.31
2	Himachal Pradesh Cricket Association Stadium, Dharamsala	10	374.00	96.13
3	Sawai Mansingh Stadium, Jaipur	21	372.81	61.88
4	M Chinnaswamy Stadium, Bengaluru	24	372.08	68.41
5	Rajiv Gandhi International Stadium, Uppal, Hyderabad	26	369.42	83.76
6	Eden Gardens, Kolkata	30	359.43	90.52
7	Wankhede Stadium, Mumbai	59	353.44	64.14
8	Narendra Modi Stadium, Ahmedabad	41	351.90	58.59
9	Brabourne Stadium	10	348.10	42.65
10	Punjab Cricket Association IS Bindra Stadium	10	347.60	26.02
11	Brabourne Stadium, Mumbai	17	342.59	69.61

Figure 12. Query 9 result: average match runs and score variation per venue (at least 10 matches). Example of the analytical SQL workflow in PostgreSQL.

7.10 Q10 - Overall team batting profile

Royal Challengers Bengaluru lead with 187.83 runs per match, a scoring rate of 168.62 runs per 100 balls and the lowest dot-ball percentage (30.72%). Gujarat Titans are close on dot balls (30.85%). Mumbai Indians (36.56%) and Kolkata Knight Riders (36.83%) play more dot balls but hit large volumes of boundaries over many more matches (for example 1,968 sixes for Mumbai).

	batting_team text	matches bigint	total_runs numeric	runs_per_match numeric	fours bigint	sixes bigint	run_rate numeric	dot_ball_percentage numeric
1	Royal Challengers Bengal...	46	8640	187.83	729	454	168.62	30.72
2	Punjab Kings	87	15509	178.26	1281	782	155.51	34.76
3	Gujarat Titans	77	13743	178.48	1239	519	153.40	30.85
4	Lucknow Super Giants	72	12754	177.14	1004	611	151.15	34.39
5	Delhi Capitals	119	19959	167.72	1798	777	144.92	34.33
6	Sunrisers Hyderabad	210	34640	164.95	2879	1333	142.59	34.09
7	Mumbai Indians	291	47641	163.71	4101	1968	141.51	36.56
8	Gujarat Lions	30	4862	162.07	460	155	141.38	33.59
9	Chennai Super Kings	265	43600	164.53	3615	1735	141.11	34.98
10	Rajasthan Royals	250	40437	161.75	3559	1564	140.89	36.33
11	Kolkata Knight Riders	278	43646	157.00	3826	1711	139.92	36.83

Figure 13. Query 10 result: runs per match, fours, sixes, scoring rate and dot-ball percentage by team.

Note on 'run rate' in Query 10: this column is computed as runs per 100 legal balls, so it is comparable to a team strike rate rather than runs per over.

8. Power BI Dashboard

The dashboard is divided into five pages. Approximate values below are read from the charts.

8.1 Page 1 - IPL Overview

The overview page has six KPI cards: Total Matches (1,243), Total Runs (402K), Total Wickets (13K), Total Sixes (16K), Total Fours (34K) and Average Innings Score (159.80). It also shows runs by season, matches by season, the top run scorers (V Kohli, RG Sharma, S Dhawan, DA Warner, AM Rahane), the leading wicket takers (YS Chahal, B Kumar, SP Narine, R Ashwin, RA Jadeja) and a toss-advantage donut chart (628 matches, 50.52%, won by the toss winner; 615, 49.48%, lost). Season, venue and team slicers filter the page.

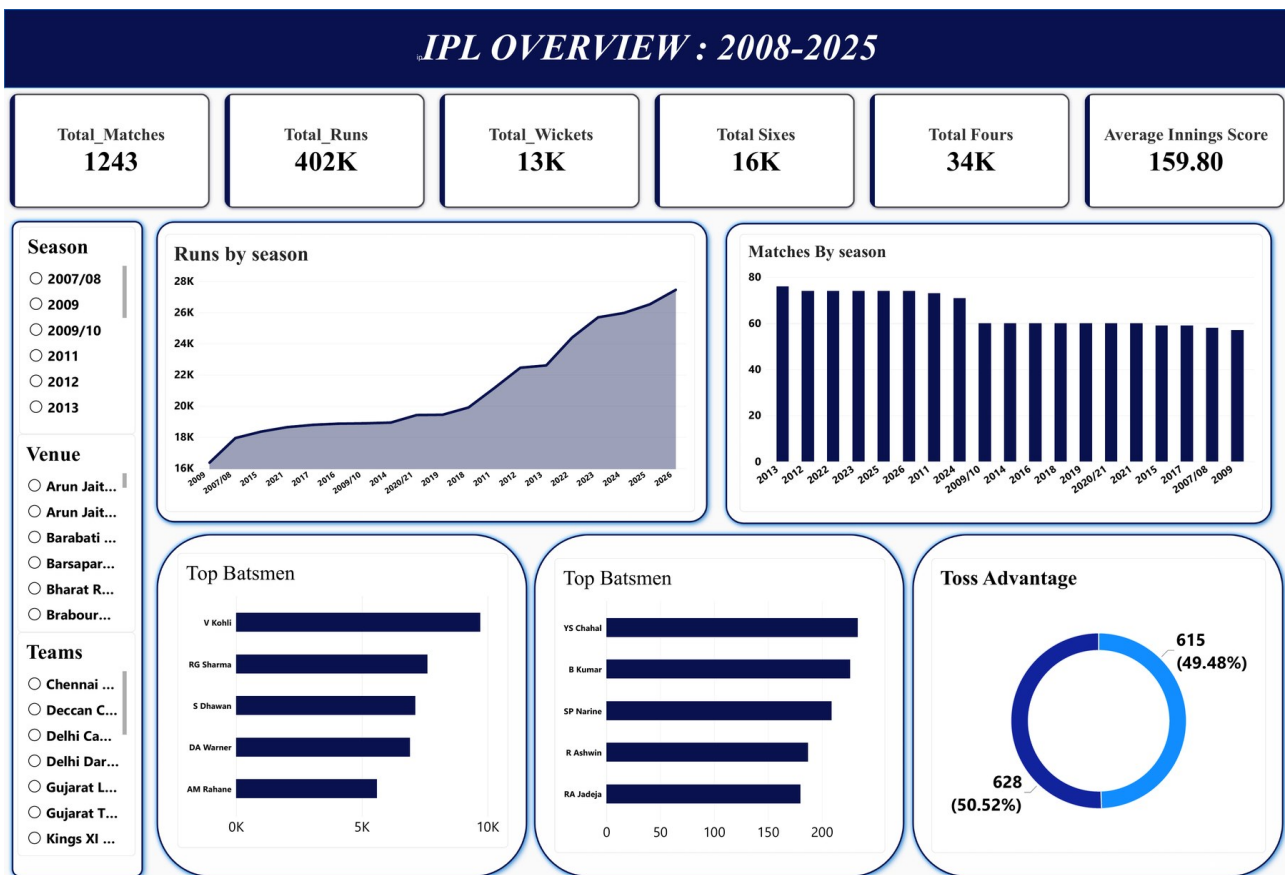


Figure 14. Page 1 - IPL Overview: headline KPIs, seasonal trends, player rankings and toss advantage.

8.2 Page 2 - Season & Scoring Trends

Six visuals show run rate, average runs per innings, average boundaries per match, total boundaries, total runs and total matches by season. Scoring has trended upward: average run rate rises from about 7.5 in 2009 to about 9.9 in 2026, and average runs per innings from about 141 to about 185. The sharpest increase begins in 2022, when the number of matches per season rose to about 74. Boundary counts rise at the same time, with sixes per match climbing from roughly 9-10 in the early seasons to about 19 in 2026. Match counts were about 73-76 in 2011-2013, about 60 from 2014 to 2021, and about 71-74 from 2022.

Season & Scoring Trends

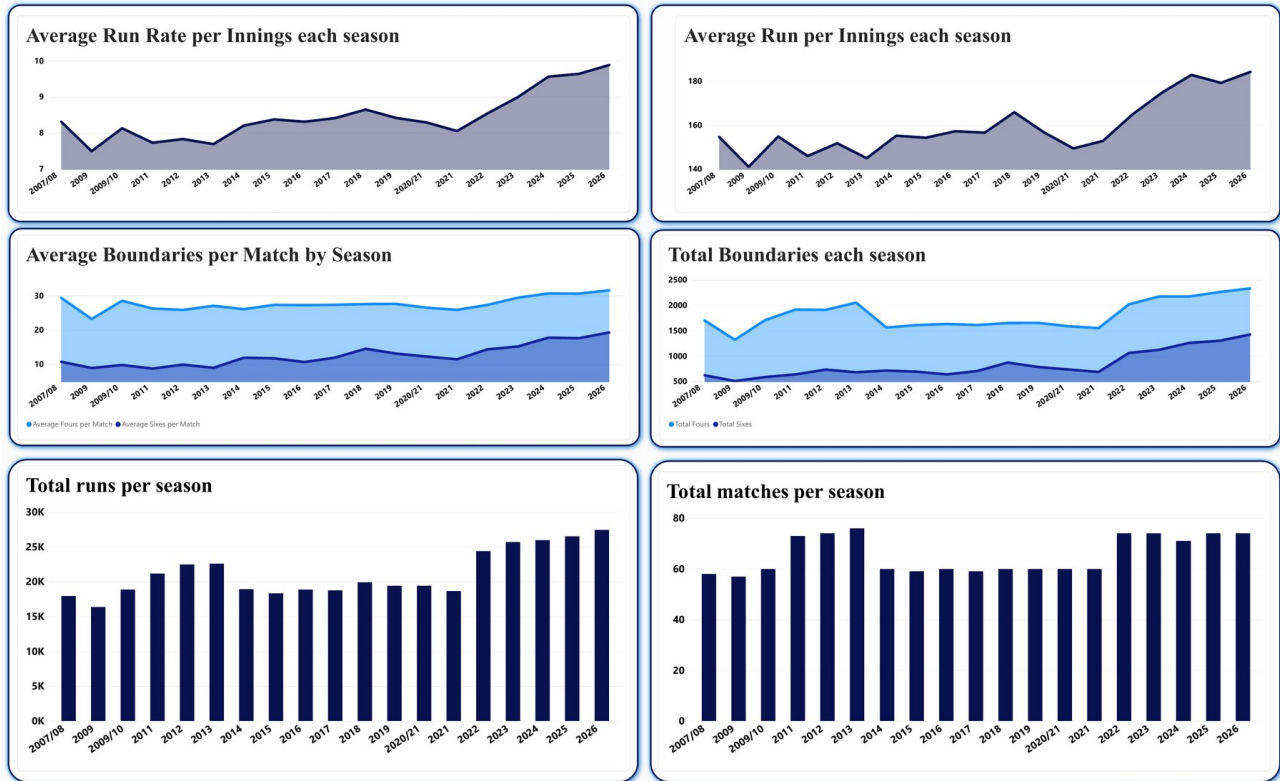


Figure 15. Page 2 - Season and scoring trends.

8.3 Page 3 - Team Performance

This page compares total wins, win percentage, successful chases, the fours-versus-sixes boundary profile, average run rate and average team runs. Mumbai Indians (about 290), Kolkata Knight Riders (about 277) and Chennai Super Kings (about 264) have the most wins. Kolkata Knight Riders (about 84) and Mumbai Indians (about 78) lead in successful chases. Royal Challengers Bengaluru top both average run rate (about 10.1) and average team runs (about 188), consistent with the SQL results. Several franchises appear under more than one name (for example Royal Challengers Bangalore and Bengaluru, Delhi Daredevils and Delhi Capitals, Kings XI Punjab and Punjab Kings), so their totals are split across entries.

Team Performance

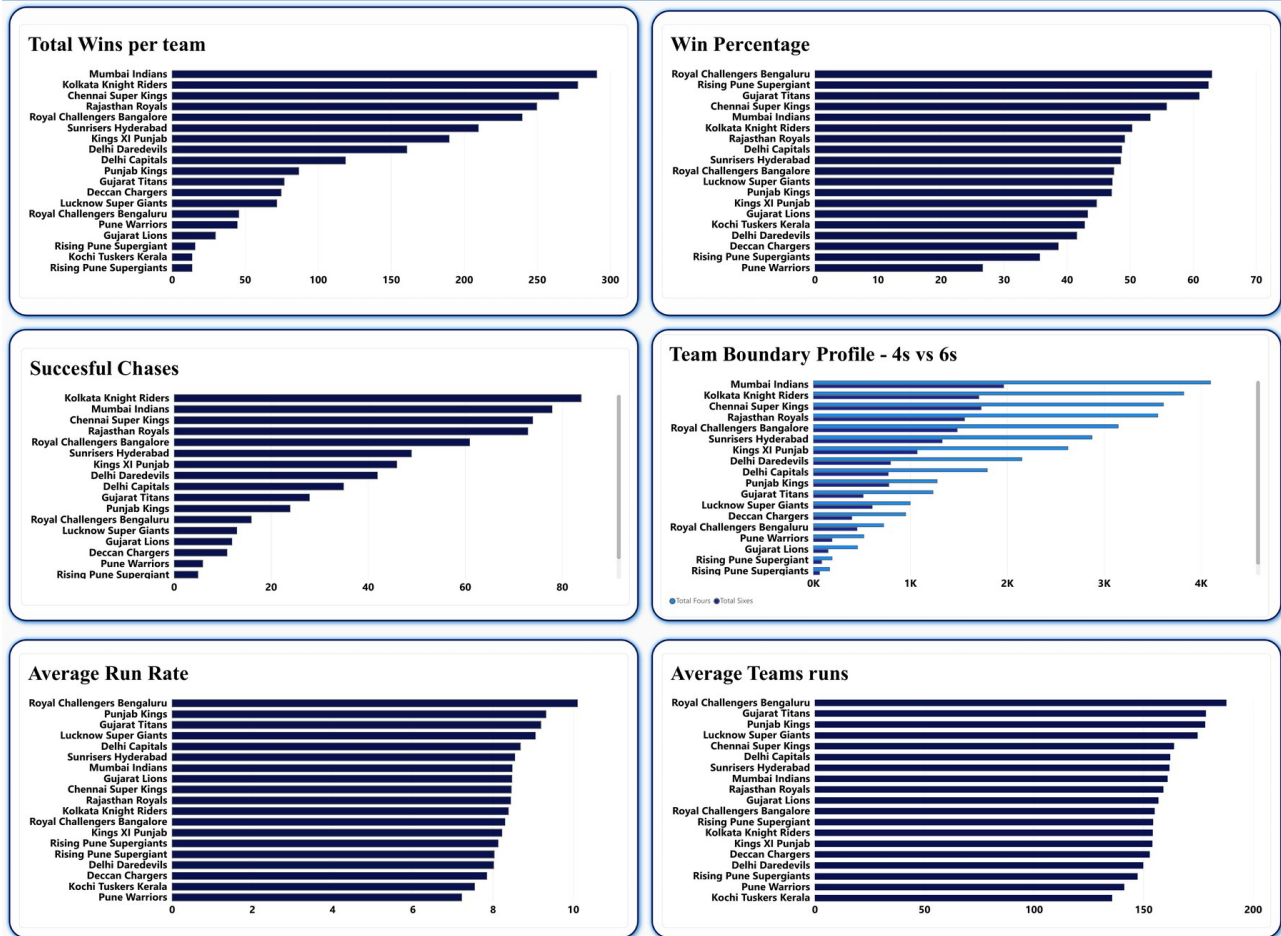


Figure 16. Page 3 - Team performance comparison.

8.4 Page 4 - Player Performance

Six ranking visuals cover top run scorers (V Kohli about 9.7K, then RG Sharma, S Dhawan, DA Warner and KL Rahul), top wicket takers (YS Chahal, B Kumar and SP Narine lead), strike rates, economy, sixes (CH Gayle about 358, RG Sharma about 323, V Kohli about 317) and fours (V Kohli about 840, DA Warner and RG Sharma about 660 each). Batter and bowler slicers allow filtering by player.

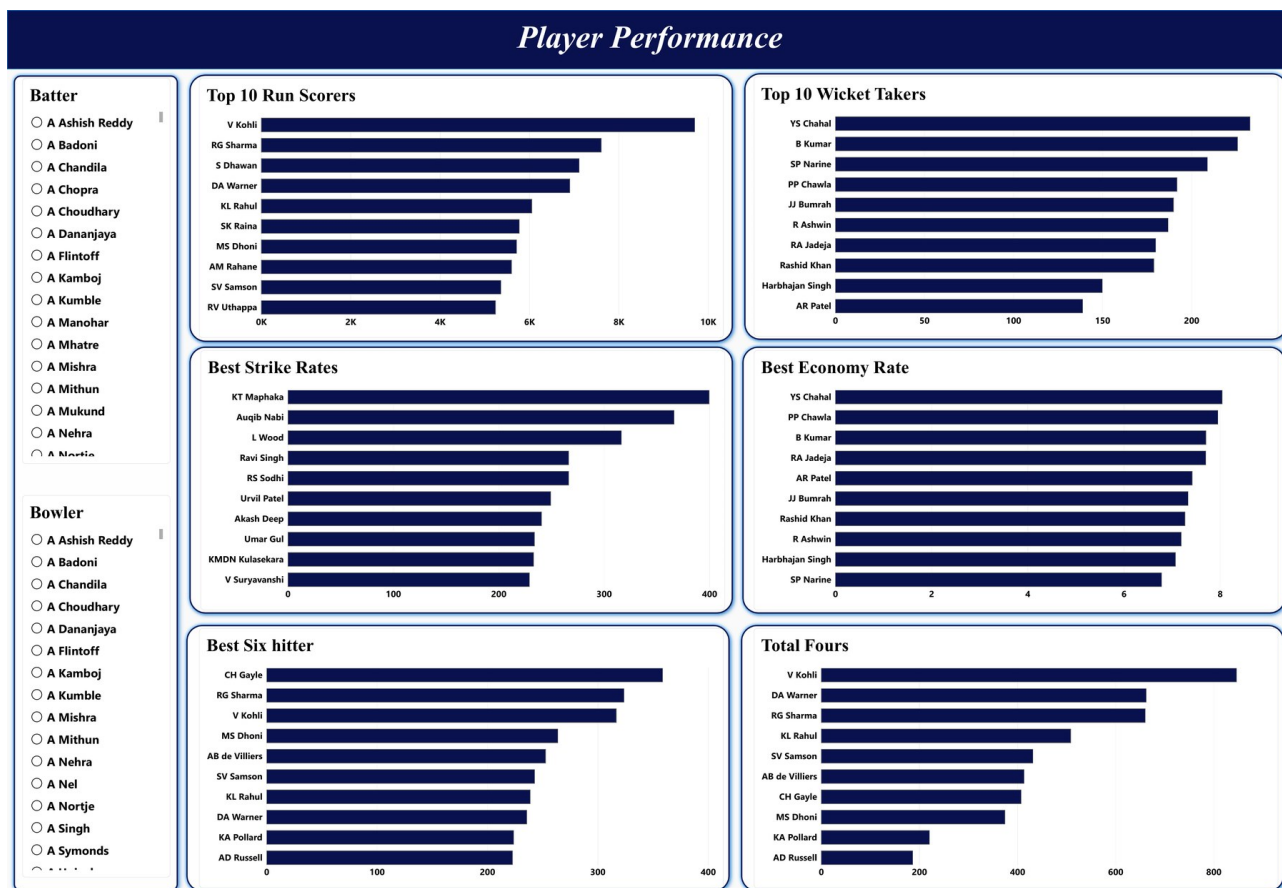


Figure 17. Page 4 - Player performance rankings.

Interpretation note: the Best Strike Rates visual shows values up to about 400 for players such as KT Maphaka and Auqib Nabi, which indicates very small samples. The reliable strike-rate ranking is the SQL result in Section 7.7, which requires at least 500 balls faced. The Best Economy Rate visual displays the economy of the leading wicket takers (about 6.8 to 8.1), not a minimum-balls ranking of the best economies in the league.

8.5 Page 5 - Venue & Match Insights

The final page shows average runs at each venue, total matches by venue, toss distribution and toss decision versus outcome. Eden Gardens (about 77 matches), Wankhede Stadium (about 73) and M Chinnaswamy Stadium (about 65) have hosted the most matches. Teams that won the toss chose to field in 825 matches (66.37%) and to bat in 418 (33.63%). In the final chart, the bar for field is about 54 and for bat about 44, consistent with the overall 50.52% toss-winner win rate once weighted by how often each decision was made. Venues such as Eden Gardens, Wankhede and Chepauk appear under more than one name, which splits their counts.

Venue & Match Insights

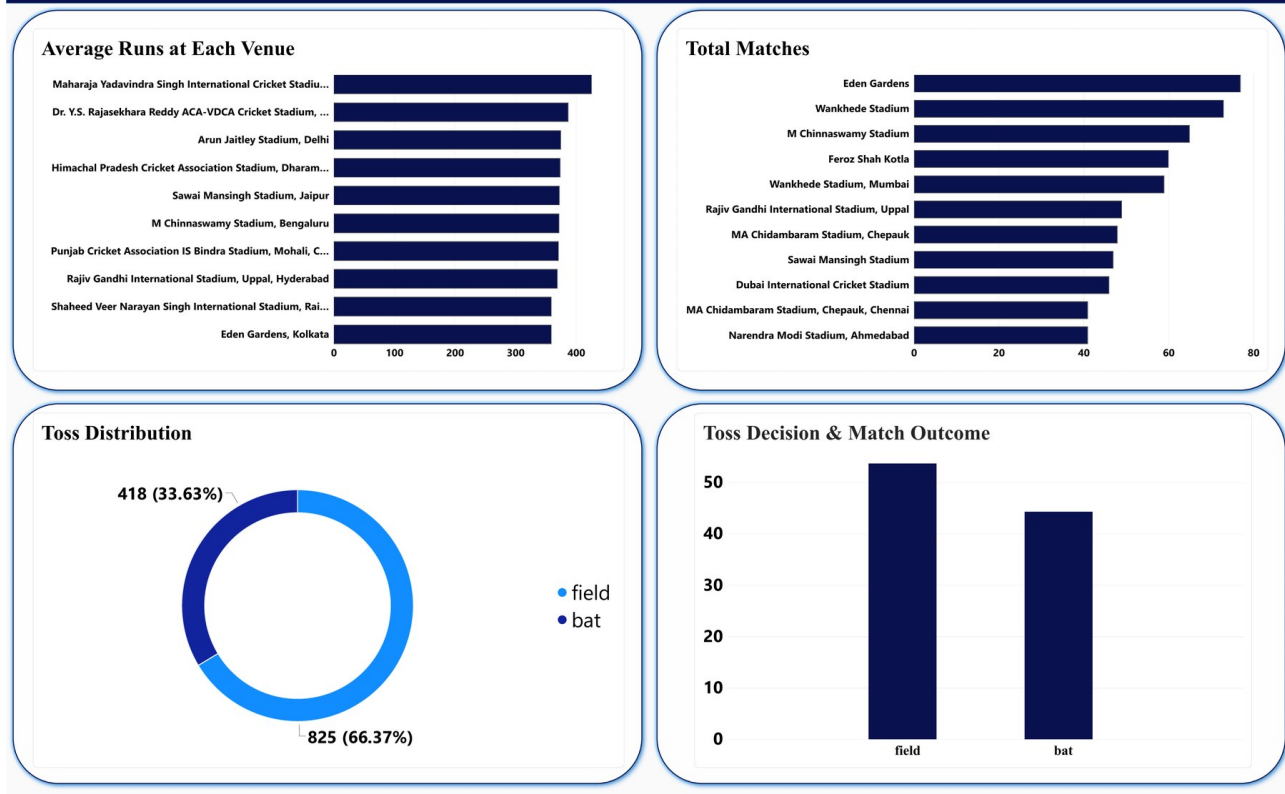


Figure 18. Page 5 - Venue and match insights.

9. Key Observations

- **Scoring has risen sharply.** Average innings score and run rate climbed from their 2009 lows (about 141 and 7.5) to about 185 and 9.9 in 2026, with the steepest growth from 2022.
- **Boundaries drive the change.** Average sixes per match roughly doubled, from about 9-10 in early seasons to about 19 in 2026.
- **Royal Challengers Bengaluru are the strongest batting side by average.** They lead in average score (187.83), Powerplay run rate (9.99) and lowest dot-ball percentage (30.72%), though from only 46 innings; Gujarat Titans and Punjab Kings follow with larger samples.
- **The toss is not decisive.** Toss winners won 50.52% of matches. Captains chose to field two times out of three, and the chart suggests field-first toss winners won more often (about 54%) than bat-first ones (about 44%), though this is descriptive and not proof of causation.
- **Venues differ widely.** Arun Jaitley Stadium, Dharamsala and Jaipur average over 370 runs per match; Wankhede, with a far larger sample, averages 353.
- **Pace bowlers dominate dot-ball pressure.** DW Steyn (46.70%), S Sreesanth and DP Nannes lead among bowlers with at least 500 legal balls.

10. Limitations

1. Results depend on the quality and completeness of the source dataset.
2. Team and venue names change over time or appear in several forms (for example Royal Challengers Bangalore/Bengaluru, Wankhede Stadium/Wankhede Stadium, Mumbai), which splits

statistics into separate categories.

3. Rate-based rankings can mislead for small samples; the SQL queries apply minimum thresholds but some dashboard visuals do not.
4. Historical comparisons are affected by changes in the number of teams and matches per season.
5. A toss decision or win margin alone does not establish causation.
6. Delivery-level measures must be aggregated at the correct match, innings, team or player level.
7. The season labels in the dataset (2007/08, 2009, 2009/10 and so on, through 2026) do not match the dashboard title '2008-2025'; the title should be updated to reflect the data.

11. Conclusion

This project used Python, Pandas, PostgreSQL, SQL and Microsoft Power BI to analyse IPL cricket data and present the results through an interactive dashboard. Data preparation gave the data a consistent structure, PostgreSQL enabled analytical queries across seasons, teams, players and venues, and Power BI organised the results into five pages for exploration.

The analysis shows that IPL batting has become markedly more aggressive, that the toss alone gives almost no advantage, and that teams and venues differ substantially in scoring profile. The project gave practical experience in data cleaning, relational database operations, SQL aggregation, measure development and dashboard design, and forms an end-to-end analytics portfolio piece.

12. Future Scope

- Add more detailed player and team comparison filters.
- Introduce match-level drill-through pages.
- Compare batting and bowling performance by innings and match phase.
- Standardise team and venue names across seasons.
- Add match-result and win-margin analysis.
- Apply minimum-sample filters to the strike-rate and economy-rate visuals so they match the SQL results.
- Publish the report to Power BI Service, subject to account and sharing permissions.

References

1. IPL ball-by-ball dataset used in the project.
2. PostgreSQL analytics report prepared for this project (IPL Analytics Using PostgreSQL).
3. Microsoft Power BI dashboard developed for IPL analysis.
4. Python and Pandas documentation.
5. PostgreSQL documentation.
6. Microsoft Power BI documentation.

Appendix A - SQL Queries

All queries run against the `ipl_data` table in PostgreSQL.

```
-- QUESTION 1
-- How has the scoring pattern of IPL changed across seasons?
-- This analysis compares the average innings score and run rate
-- for every season. It helps us understand whether batting has
-- become more aggressive over the years.
WITH innings_data AS (
    SELECT
        season,
        match_id,
        innings,
        SUM(runs_total) AS innings_runs,
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls
    FROM ipl_data
    GROUP BY season, match_id, innings
)
SELECT
    season,
    COUNT(DISTINCT match_id) AS matches,
    ROUND(AVG(innings_runs), 2) AS average_innings_score,
    ROUND(
        SUM(innings_runs) * 6.0 /
        NULLIF(SUM(legal_balls), 0),
        2
    ) AS run_rate
FROM innings_data
GROUP BY season
ORDER BY season;
```

```
-- QUESTION 2
-- Which teams are the most consistent at scoring runs?
-- Instead of looking only at total runs, this analysis looks at
-- average score, highest score, lowest score and score variation.
-- A lower variation means the team scores more consistently.
WITH team_innings AS (
    SELECT
        batting_team,
        match_id,
        innings,
        SUM(runs_total) AS total_runs
    FROM ipl_data
    GROUP BY batting_team, match_id, innings
)
SELECT
    batting_team,
    COUNT(*) AS innings_played,
    ROUND(AVG(total_runs), 2) AS average_score,
    ROUND(STDDEV(total_runs), 2) AS score_variation,
    MIN(total_runs) AS lowest_score,
    MAX(total_runs) AS highest_score
FROM team_innings
GROUP BY batting_team
HAVING COUNT(*) >= 20
ORDER BY average_score DESC;
```

```

-- QUESTION 3
-- Which teams are the most successful when chasing a target?
-- This compares the first innings score with the second innings
-- score and calculates the percentage of successful chases.
-- It helps identify teams that perform well under chasing pressure.
WITH innings_scores AS (
    SELECT
        match_id,
        innings,
        batting_team,
        SUM(runs_total) AS runs
    FROM ipl_data
    GROUP BY match_id, innings, batting_team
),
match_scores AS (
    SELECT
        match_id,
        MAX(
            CASE
                WHEN innings = 1 THEN runs
            END
        ) AS first_innings_score,
        MAX(
            CASE
                WHEN innings = 2 THEN batting_team
            END
        ) AS chasing_team,
        MAX(
            CASE
                WHEN innings = 2 THEN runs
            END
        ) AS chasing_score
    FROM innings_scores
    GROUP BY match_id
)
SELECT
    chasing_team,
    COUNT(*) AS chases,
    COUNT(*) FILTER (
        WHERE chasing_score > first_innings_score
    ) AS successful_chases,
    ROUND(
        COUNT(*) FILTER (
            WHERE chasing_score > first_innings_score
        ) * 100.0 / COUNT(*),
        2
    ) AS chase_success_rate
FROM match_scores
WHERE chasing_team IS NOT NULL
GROUP BY chasing_team
HAVING COUNT(*) >= 10
ORDER BY chase_success_rate DESC;

```

```

-- QUESTION 4
-- Does winning the toss actually increase the chance of winning
-- the match?
-- This analysis compares the toss winner with the actual match
-- winner. It tells us whether the toss provides a meaningful
-- historical advantage.
WITH match_results AS (
    SELECT DISTINCT
        match_id,
        toss_winner,
        match_won_by
    FROM ipl_data
    WHERE toss_winner IS NOT NULL
        AND match_won_by IS NOT NULL
)
SELECT
    COUNT(*) AS total_matches,
    COUNT(*) FILTER (
        WHERE toss_winner = match_won_by
    ) AS toss_winner_wins,
    ROUND(
        COUNT(*) FILTER (
            WHERE toss_winner = match_won_by
        ) * 100.0 / COUNT(*),
        2
    ) AS toss_win_percentage
FROM match_results;

```

```

-- QUESTION 5
-- Which teams score most aggressively during the Powerplay?
-- The first six overs can strongly influence an innings.
-- This query calculates the average Powerplay score and run rate
-- for each team, allowing us to compare their early-innings
-- attacking ability.
WITH powerplay AS (
    SELECT
        batting_team,
        match_id,
        SUM(runs_total) AS runs,
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls
    FROM ipl_data
    WHERE over BETWEEN 0 AND 5
    GROUP BY batting_team, match_id
)
SELECT
    batting_team,
    COUNT(*) AS innings,
    ROUND(
        AVG(runs),
        2
    ) AS average_powerplay_runs,
    ROUND(
        SUM(runs) * 6.0 /
        NULLIF(SUM(legal_balls), 0),
        2
    ) AS powerplay_run_rate
FROM powerplay
GROUP BY batting_team
HAVING COUNT(*) >= 20
ORDER BY powerplay_run_rate DESC;

```

```

-- QUESTION 6
-- Which teams accelerate the most from the Powerplay to the
-- Death Overs?
-- This compares the run rate during overs 0-5 with the run rate
-- during overs 16-19. A high acceleration value indicates that
-- a team becomes significantly more aggressive at the end
-- of its innings.
WITH phase_scores AS (
    SELECT
        batting_team,
        match_id,
        SUM(
            CASE
                WHEN over BETWEEN 0 AND 5
                THEN runs_total
                ELSE 0
            END
        ) AS powerplay_runs,
        SUM(
            CASE
                WHEN over BETWEEN 16 AND 19
                THEN runs_total
                ELSE 0
            END
        ) AS death_runs,
        COUNT(*) FILTER (
            WHERE over BETWEEN 0 AND 5
            AND valid_ball = 1
        ) AS powerplay_balls,
        COUNT(*) FILTER (
            WHERE over BETWEEN 16 AND 19
            AND valid_ball = 1
        ) AS death_balls
    FROM ipl_data
    GROUP BY batting_team, match_id
)
SELECT
    batting_team,
    ROUND(
        SUM(powerplay_runs) * 6.0 /
        NULLIF(SUM(powerplay_balls), 0),
        2
    ) AS powerplay_run_rate,
    ROUND(
        SUM(death_runs) * 6.0 /
        NULLIF(SUM(death_balls), 0),
        2
    ) AS death_run_rate,
    ROUND(
        (
            SUM(death_runs) * 6.0 /
            NULLIF(SUM(death_balls), 0)
        )
        -
        (
            SUM(powerplay_runs) * 6.0 /
            NULLIF(SUM(powerplay_balls), 0)
        ),
        2
    ) AS acceleration
FROM phase_scores
GROUP BY batting_team
ORDER BY acceleration DESC;

```

```

-- QUESTION 7
-- Which batsmen have the best strike rate while maintaining
-- a meaningful sample size?
-- Ranking only by strike rate can be misleading because a player
-- with very few balls may appear at the top. Here we consider
-- only batsmen who have faced at least 500 legal balls.
WITH batting AS (
    SELECT
        batter,
        SUM(runs_batter) AS runs,
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS balls
    FROM ipl_data
    GROUP BY batter
)
SELECT
    batter,
    runs,
    balls,
    ROUND(
        runs * 100.0 /
        NULLIF(balls, 0),
        2
    ) AS strike_rate
FROM batting
WHERE balls >= 500
ORDER BY strike_rate DESC
LIMIT 20;

```

```

-- QUESTION 8
-- Which bowlers create the most dot-ball pressure?
-- Wickets are not the only measure of bowling quality.
-- Dot balls create pressure and can force batsmen to take
-- risky shots. This query compares dot-ball percentage among
-- bowlers with a reasonable number of deliveries.
SELECT
    bowler,
    COUNT(*) FILTER (
        WHERE valid_ball = 1
    ) AS legal_balls,
    COUNT(*) FILTER (
        WHERE valid_ball = 1
        AND runs_total = 0
    ) AS dot_balls,
    ROUND(
        COUNT(*) FILTER (
            WHERE valid_ball = 1
            AND runs_total = 0
        ) * 100.0 /
        NULLIF(
            COUNT(*) FILTER (
                WHERE valid_ball = 1
            ),
            0
        ),
        2
    ) AS dot_ball_percentage
FROM ipl_data
GROUP BY bowler
HAVING COUNT(*) FILTER (
    WHERE valid_ball = 1
) >= 500
ORDER BY dot_ball_percentage DESC;

```

```

-- QUESTION 9
-- Which venues are the most batting-friendly?
-- Different stadiums can produce very different scoring patterns.
-- This query calculates average match runs at each venue so that
-- we can identify venues where high-scoring matches are more common.
WITH venue_matches AS (
    SELECT
        venue,
        match_id,
        SUM(runs_total) AS total_runs
    FROM ipl_data
    GROUP BY venue, match_id
)
SELECT
    venue,
    COUNT(*) AS matches,
    ROUND(
        AVG(total_runs),
        2
    ) AS average_match_runs,
    ROUND(
        STDDEV(total_runs),
        2
    ) AS score_variation
FROM venue_matches
GROUP BY venue
HAVING COUNT(*) >= 10
ORDER BY average_match_runs DESC;

```

```

-- QUESTION 10
-- Which teams have the strongest overall batting profile?
-- This combines multiple batting indicators instead of ranking
-- teams using only total runs. It compares runs per match,
-- run rate, boundaries and dot-ball percentage to understand
-- how teams build their innings.
WITH team_stats AS (
    SELECT
        batting_team,
        SUM(runs_total) AS total_runs,
        COUNT(DISTINCT match_id) AS matches,
        COUNT(*) FILTER (
            WHERE runs_batter = 4
        ) AS fours,
        COUNT(*) FILTER (
            WHERE runs_batter = 6
        ) AS sixes,
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls,
        COUNT(*) FILTER (
            WHERE runs_total = 0
            AND valid_ball = 1
        ) AS dot_balls
    FROM ipl_data
    GROUP BY batting_team
)
SELECT
    batting_team,
    matches,
    total_runs,
    ROUND(
        total_runs * 1.0 /
        NULLIF(matches, 0),
        2
    ) AS runs_per_match,
    fours,
    sixes,
    ROUND(
        total_runs * 100.0 /
        NULLIF(legal_balls, 0),
        2
    ) AS run_rate,
    ROUND(
        dot_balls * 100.0 /
        NULLIF(legal_balls, 0),
        2
    ) AS dot_ball_percentage
FROM team_stats
ORDER BY run_rate DESC;

```

